



Polar Coordinates · Worksheet 1

Core Pure 2 Chapter 5: Polar Coordinates. Answer all questions. Show your working clearly. Total: 65 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 5. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Angles are in radians. Polar coordinates (r, θ) have $r \geq 0$ and $-\pi < \theta \leq \pi$ unless stated otherwise; curves are only drawn where $r \geq 0$. Give exact answers where possible.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Converting between polar and Cartesian forms	19	
B Sketching polar curves	21	
C Area enclosed by a polar curve	20	
Roar-level challenge	5	
Total	65	

Section A Converting between polar and Cartesian forms

19 marks

1. Find the polar coordinates of the point with Cartesian coordinates $(-3, 3\sqrt{3})$. [2]

.....

.....

2. Find the exact Cartesian coordinates of the point with polar coordinates $(4, -\frac{3\pi}{4})$. [2]

.....

.....



3. Show that the curve with Cartesian equation $(x^2 + y^2)^2 = 4xy$ has polar equation $r^2 = 2 \sin 2\theta$. [3]

.....

.....

.....

.....

4. Find a Cartesian equation of the curve with polar equation [4]

$$r = \frac{2}{1 - \cos \theta}, \quad 0 < \theta < 2\pi,$$

giving your answer in the form $y^2 = f(x)$.

.....

.....

.....

.....

.....



5. The curve C has polar equation $r = 4 \cos \theta$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

(a) Find a Cartesian equation of C , and hence show that C is a circle, stating its centre and radius. [3]

.....

.....

.....

.....

(b) The line l has Cartesian equation $x = 3$. Find a polar equation of l . [2]

.....

.....

(c) Find the polar coordinates of the points where l meets C . [3]

.....

.....

.....

.....

(Total for Question 5 is 8 marks)

**Section B Sketching polar curves**

21 marks

6. On the same diagram, sketch the curves with polar equations $r = 3$ and $\theta = \frac{3\pi}{4}$. [2]

7. Sketch the curve with polar equation $r = 2\theta$, $0 \leq \theta \leq 2\pi$. [3]
Label the polar coordinates of the points where the curve meets the half-lines $\theta = \pi$ and $\theta = 2\pi$ (the initial line).

8. Sketch the curve with polar equation $r = 4 + \cos \theta$, $-\pi < \theta \leq \pi$. [3]
Label the points where the curve meets the initial line, the half-line $\theta = \pi$ and the half-lines $\theta = \pm \frac{\pi}{2}$.

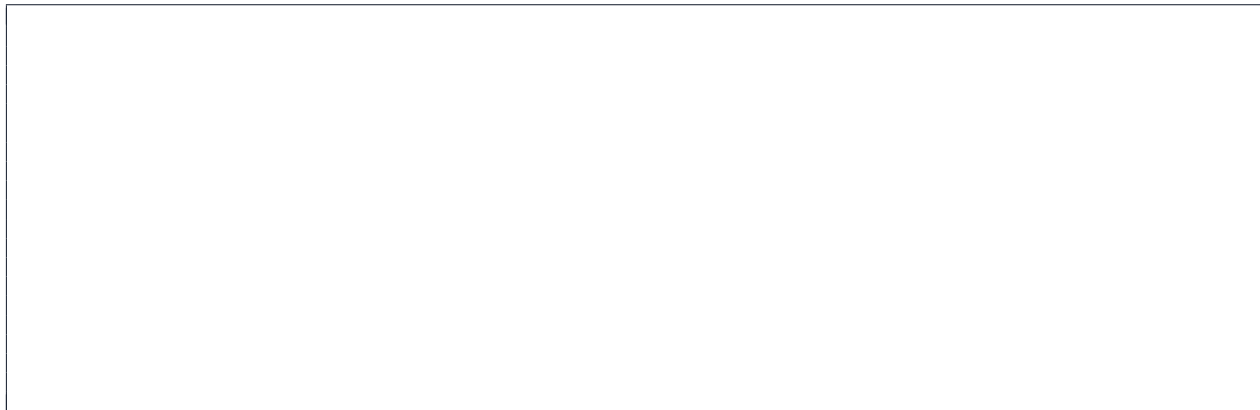


9. The curve C has polar equation $r = 2 \sin 3\theta$, where $r \geq 0$ and $-\pi < \theta \leq \pi$.

[4]

(a) Find the intervals of θ for which the curve exists.

(b) Sketch C , labelling the polar coordinates of the point of each loop that is furthest from the pole.





10. The curve C has polar equation

$$r = 2(1 + \cos \theta), \quad -\pi < \theta \leq \pi.$$

(a) Sketch C , labelling the point where C meets the initial line.

[2]

(b) Find the polar coordinates of the points on C , other than the pole, where the tangent to C is parallel to the initial line.

[4]

.....

.....

.....

.....

.....

(c) Find the polar coordinates of the points on C , other than the pole, where the tangent to C is perpendicular to the initial line.

[3]

.....

.....

.....

.....

(Total for Question 10 is 9 marks)

**Section C Area enclosed by a polar curve**

20 marks

11. Find the exact area of the region bounded by the curve $r = 2\theta$, $0 \leq \theta \leq \pi$, and the half-line $\theta = \pi$. [2]

.....

.....

12. Find the exact area enclosed by one loop of the curve $r = 2\sin 3\theta$ (see Question 9). [3]

.....

.....

.....

.....

13. Find the exact area enclosed by the curve $r = 4 + \cos \theta$, $-\pi < \theta \leq \pi$. [3]

.....

.....

.....

.....

14. The curve C has polar equation $r = 1 + \tan \theta$, $0 \leq \theta \leq \frac{\pi}{4}$. [4]
Show that the area of the region bounded by C , the initial line and the half-line $\theta = \frac{\pi}{4}$ is $\frac{1}{2}(1 + \ln 2)$.

.....

.....

.....

.....

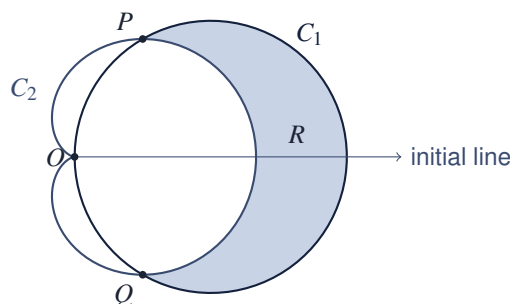
.....



15. The diagram shows the curves C_1 and C_2 with polar equations

$$C_1 : r = 3 \cos \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \quad C_2 : r = 1 + \cos \theta, \quad -\pi < \theta \leq \pi.$$

The curves meet at the points P and Q . The shaded region R lies inside C_1 and outside C_2 .



(a) Show that P and Q have polar coordinates $(\frac{3}{2}, \frac{\pi}{3})$ and $(\frac{3}{2}, -\frac{\pi}{3})$. [2]

.....

.....

(b) Show that the area of R is π . [6]

.....

.....

.....

.....

.....

.....

.....

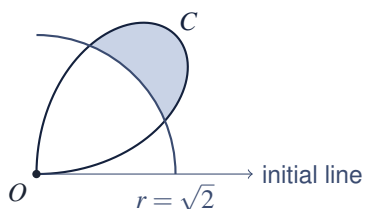
(Total for Question 15 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. The diagram shows the curve C with polar equation $r = 2 \sin 2\theta$, $0 \leq \theta \leq \frac{\pi}{2}$, and the circle with polar equation $r = \sqrt{2}$. The shaded region lies inside C and outside the circle. Find the exact area of the shaded region. [5]



.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

END OF WORKSHEET

Total: 65 marks